

其中 $R(t) = A \sin \omega t$ $R(s) = \frac{A\omega}{s^2 + \omega^2} = \frac{A\omega}{(s + j\omega)(s - j\omega)}$

系统输出的**稳态分量**为 $c(t)_w = be^{-j\omega t} + \bar{b}e^{j\omega t}$

$$\begin{aligned}
 c(t)_w &= -|G(j\omega)|e^{-j\phi(\omega)} \frac{Ae^{-j\omega t}}{2j} + |G(j\omega)|e^{j\phi(\omega)} \frac{Ae^{j\omega t}}{2j} \\
 &= |G(j\omega)|A \frac{e^{j(\omega t + \phi)} - e^{-j(\omega t + \phi)}}{2j} = |G(j\omega)|A \sin(\omega t + \phi)
 \end{aligned}$$

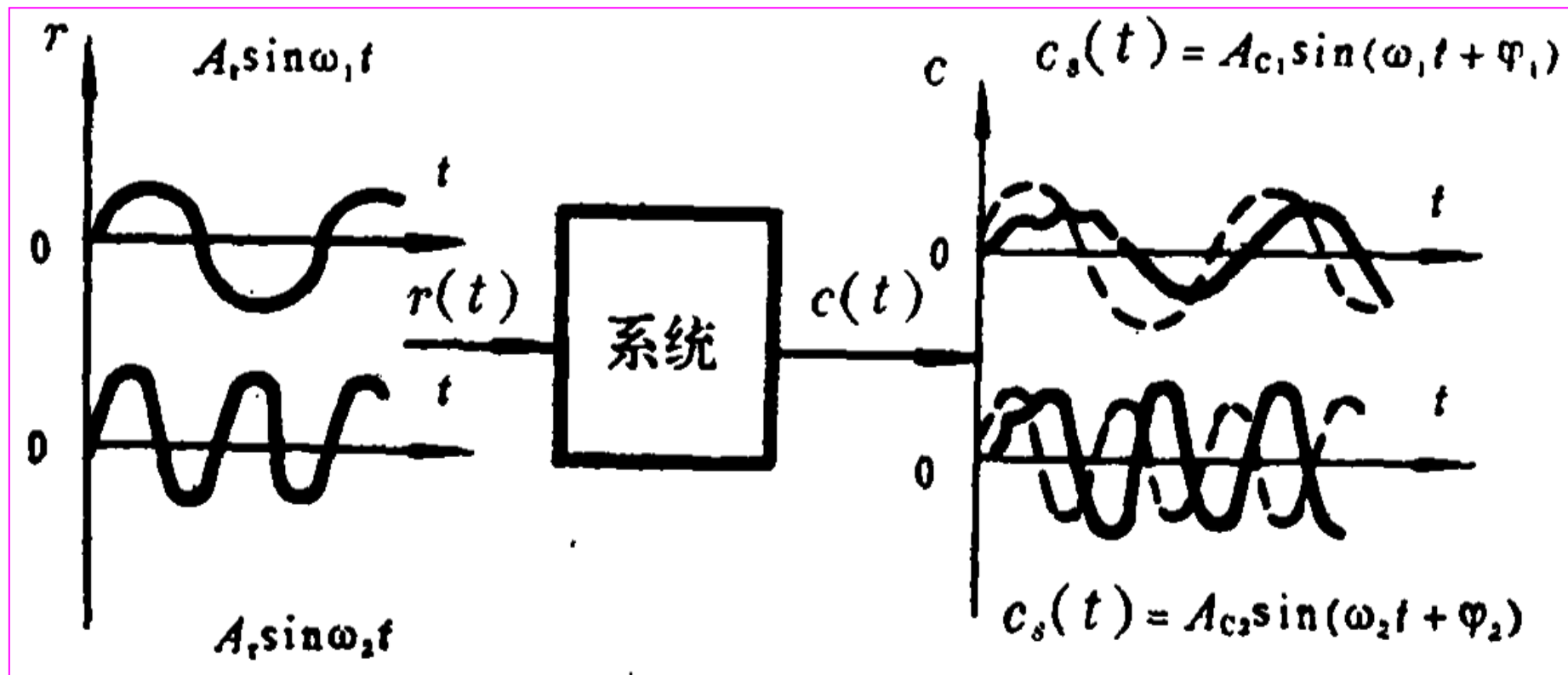
$$G(j\omega) = G(s)\big|_{s=j\omega} = |G(j\omega)|e^{j\angle G(j\omega)}$$

$$\left. \begin{array}{l} |G(j\omega)| \text{ 称为幅频特性} \\ \angle G(j\omega) = \arctg \frac{\operatorname{Im} G(j\omega)}{\operatorname{Re} G(j\omega)} \text{ 称为相频特性} \end{array} \right\} \begin{array}{l} \text{通称为系统的频率特性} \\ \text{它们都是 } \omega \text{ 的函数} \end{array}$$

✦ 注意：

- $|G(j\omega)|$ 反应了系统在不同频率的正弦输入信号作用下，**稳态**输出的幅值和输入信号**幅值之比**。
- $\angle G(j\omega)$ 反应了系统在不同频率的正弦输入信号作用下，**稳态**输出信号相对于输入信号的**相位位移**。

物理意义

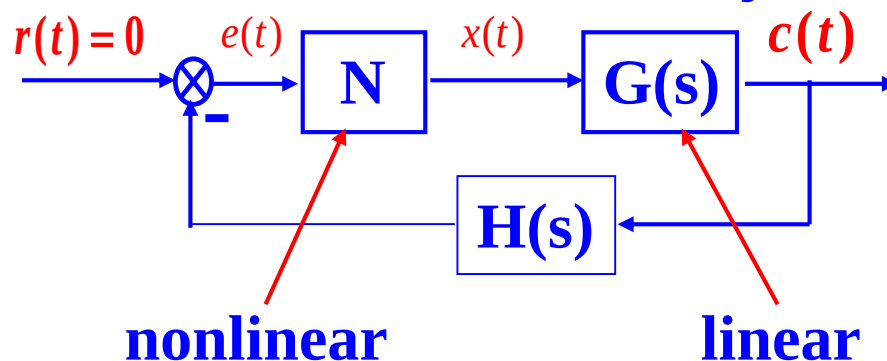


§ 8.3 Describing function method

- Describing function method is first proposed by *P.J. Daniel* in 1940.

Basic idea

For the nonlinear system



Typical structure of the nonlinear systems

If : $e(t) = A \sin \omega t \longrightarrow$ a sinusoidal input,
 $x(t)$, maybe it is not a sinusoidal but a periodic function, can be expressed as a Fourier series.

Assumption:

The harmonic of $x(t)$ could be neglected, then:

$x(t) \approx x_1 \sin(\omega t + \phi_1) \Rightarrow$ output frequency is equal to input frequency approximately.

Similar to the *frequency analysis* of linear system, we can perform frequency analysis for the nonlinear system based on the assumption.

$\Rightarrow$ Describing function

§ 8.3 Describing function method

- The describing function method is mainly used to analyze the *stability* and *self-oscillation* of the nonlinear system without external excitation.
- Advantage:
 - It is not confined by *the order of the system*.
- Disadvantages:
 - It is an *approximate analysis method*.
 - It can only be used to study the system *frequency characteristics*.

8.3.1 Concept of describing function

The describing function method can be applied to nonlinear systems with the following features:

1. The linear part and the nonlinear part can be separated.

Shown in Fig. 8-9, NL is a *nonlinear part*, G is the transfer function of the *linear part*.

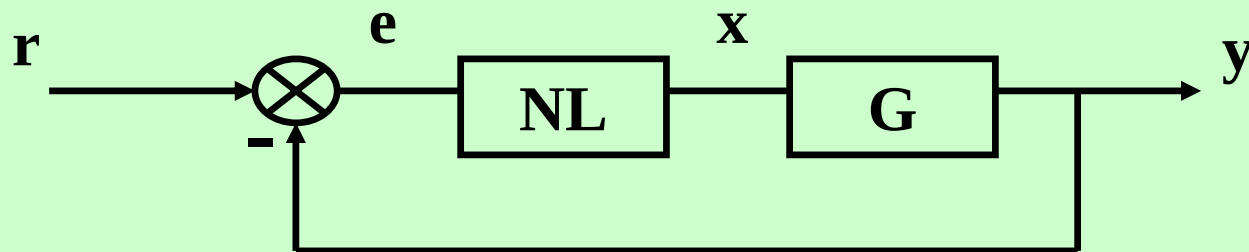


Fig. 8-9 Typical structure of nonlinear system

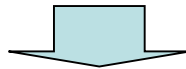
2. The system has an *odd-symmetric nonlinearity*, and the input-output relationship of the nonlinear part is static (without energy storage elements).

- If so, sinusoidal input $\rightarrow$ periodic output
- The output can be expanded into Fourier series with a *zero D. C. component*.

$$f(x) = -f(-x)$$

3. The linear part is a good low-pass filter

We can suppose the *higher-order harmonic* is filtered out.



There is only a *fundamental component* in the the output .

If all the conditions above are satisfied, we can describe the nonlinear components by the frequency response like as that we did in the linear systems.

So we have:

Definition of the describing function

The describing function $N(A)$ of the nonlinear element is: the *complex ratio* of the fundamental component of the output $y(t)$ and the sinusoidal input $e(t)$, that is:

For $e(t) = A \sin \omega t$,

$$\begin{aligned} x(t) &\approx A_1 \cos \omega t + B_1 \sin \omega t \\ &= x_1 \sin(\omega t + \phi_1) \quad \Longrightarrow \quad N(A) = \frac{x_1 e^{j\phi_1}}{A} \end{aligned}$$

Assume the input of nonlinear is sinusoidal $e(t) = A \sin \omega t$

Normally, the output is periodic, which can be expressed as a *Fourier series*:

$$x(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} (A_n \cos n\omega t + B_n \sin n\omega t)$$

The nonlinearity is *odd-symmetric*

$$\Rightarrow A_0 = 0$$

$$A_n = \frac{1}{\pi} \int_0^{2\pi} x(t) \cos n\omega t \, d(\omega t)$$

$$B_n = \frac{1}{\pi} \int_0^{2\pi} x(t) \sin n\omega t \, d(\omega t)$$

For the fundamental component, we have

$$A_1 = \frac{1}{\pi} \int_0^{2\pi} x(t) \cos \omega t \, d(\omega t)$$

$$B_1 = \frac{1}{\pi} \int_0^{2\pi} x(t) \sin \omega t \, d(\omega t)$$

Thus, the fundamental component is

$$x_1(t) = A_1 \cos \omega t + B_1 \sin \omega t = x_1 \sin(\omega t + \phi_1)$$

where

$$x_1 = \sqrt{A_1^2 + B_1^2} \quad \phi_1 = \arctg \frac{A_1}{B_1}$$

The describing function is then given by

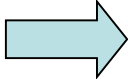
$$N(A) = \frac{x_1}{A} e^{j\phi_1}$$

Obviously, the describing function is a function of the input amplitude A . So we can regard it as a *variable gain amplifier*.

$$N(A) = \frac{\sqrt{A_1^2 + B_1^2}}{A} e^{j \arctg \frac{A_1}{B_1}} = \frac{B_1}{A} + j \frac{A_1}{A}$$

Replacing the nonlinear part by $N(A)$, we can extend the *frequency response method* of linear system to the nonlinear system so as to analyze the *frequency characteristics* of nonlinear system.

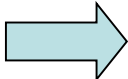
Remarks:

If the nonlinearity is *single-valued odd-symmetric* 

The output $x(t)$ is *an odd function*.

$$A_1 = 0 \quad N(A) = B_1/A$$

The describing function is a *real function* of input amplitude A .

If the nonlinearity is not *single-valued odd-symmetric* 

The output $x(t)$ is *neither an odd nor even*.

$$A_1 \neq 0, B_1 \neq 0$$

The describing function is a *complex function* of input amplitude A .

➤ Questions:

- ① How to calculate the describing function for a given nonlinear component?

Tasks:

- Go to the SPOC course (7')

8.04 典型非线性描述函数的求取

8.3.2 Describing function of typical nonlinear characteristic

1. Saturation

Assume the input is $e(t) = A \sin \omega t$

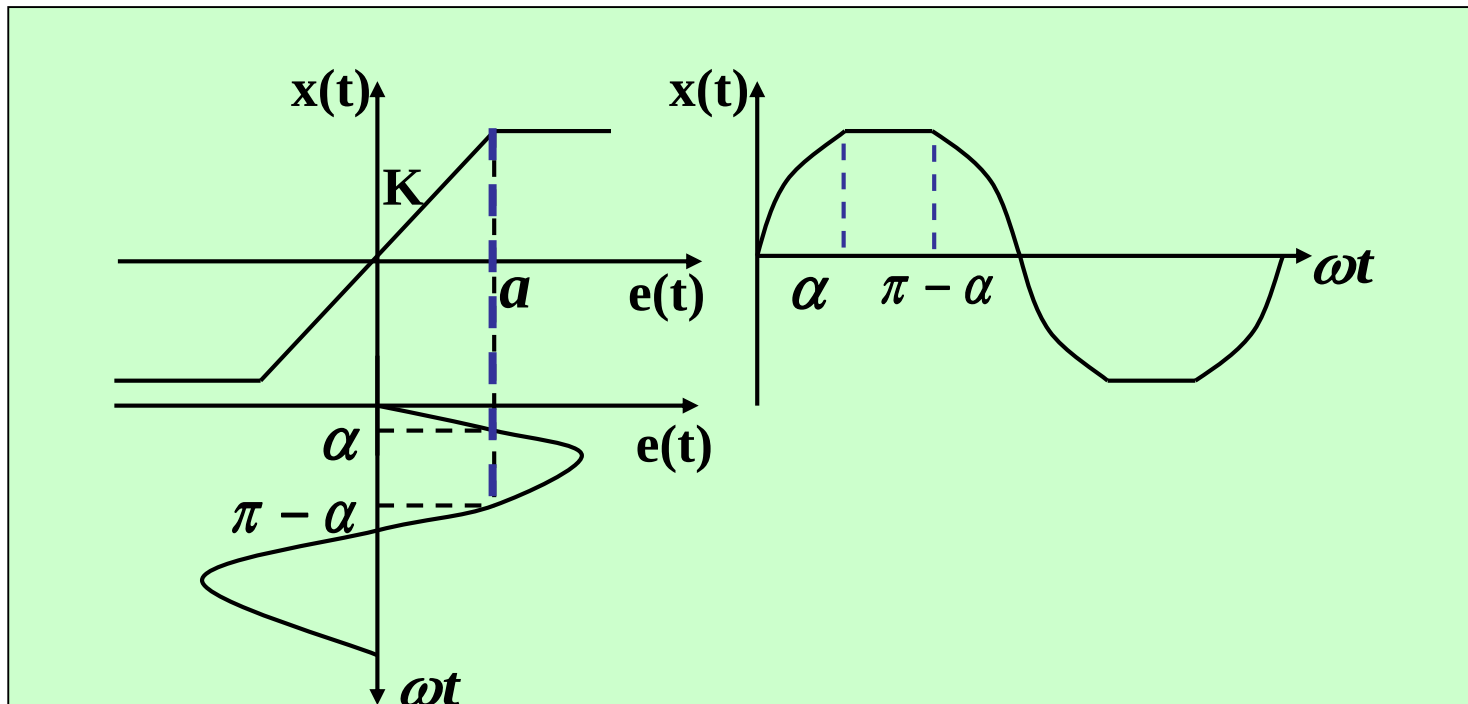


Fig 8-10 Input and output waveform of a saturation nonlinearity

when $A > a$, the output $x(t)$ is

$$x(t) = \begin{cases} KA \sin \omega t, & 0 \leq \omega t \leq \alpha \\ Ka, & \alpha < \omega t \leq \pi - \alpha \\ KA \sin \omega t, & \pi - \alpha < \omega t \leq \pi \end{cases}$$

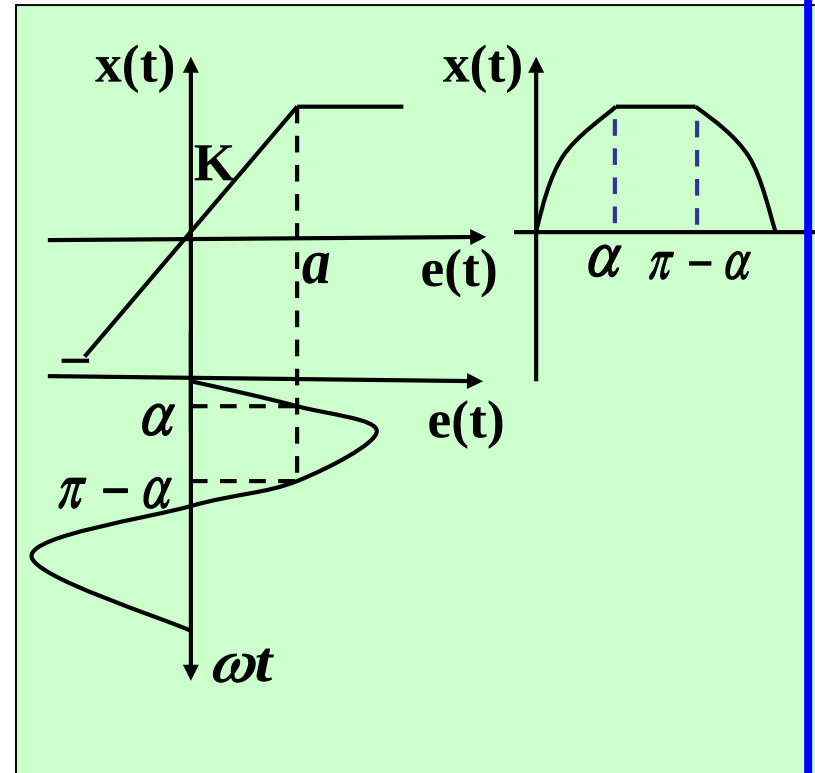
where,

$$A \sin \alpha = a, \therefore \alpha = \sin^{-1} \frac{a}{A}$$

Because the output is odd,

$$A_1 = 0 \text{ (single-valued odd-symmetric)}$$

$$\phi_1 = \tan^{-1} \frac{A_1}{B_1} = 0$$



$$\begin{aligned}
 B_1 &= \frac{2}{\pi} \int_0^\pi x(t) \sin \omega t \, d(\omega t) \\
 &= \frac{2}{\pi} \left[\int_0^\alpha KA \sin^2 \omega t \, d(\omega t) + \int_\alpha^{\pi-\alpha} Ka \sin \omega t \, d(\omega t) + \int_{\pi-\alpha}^\pi KA \sin^2 \omega t \, d(\omega t) \right] \\
 &= \frac{2}{\pi} KA \left[\sin^{-1} \frac{a}{A} + \frac{a}{A} \sqrt{1 - \left(\frac{a}{A} \right)^2} \right]
 \end{aligned}$$

The describing function of saturation nonlinearity is:

$$N(A) = \frac{B_1}{A} = \frac{2}{\pi} K \left[\sin^{-1} \frac{a}{A} + \frac{a}{A} \sqrt{1 - \left(\frac{a}{A} \right)^2} \right]$$

$N(A)$ is a *nonlinear real function* of input amplitude A .
It can be regarded as a *variable gain amplifier*.

2. Dead zone

Assume the input is $e(t) = A \sin \omega t$,

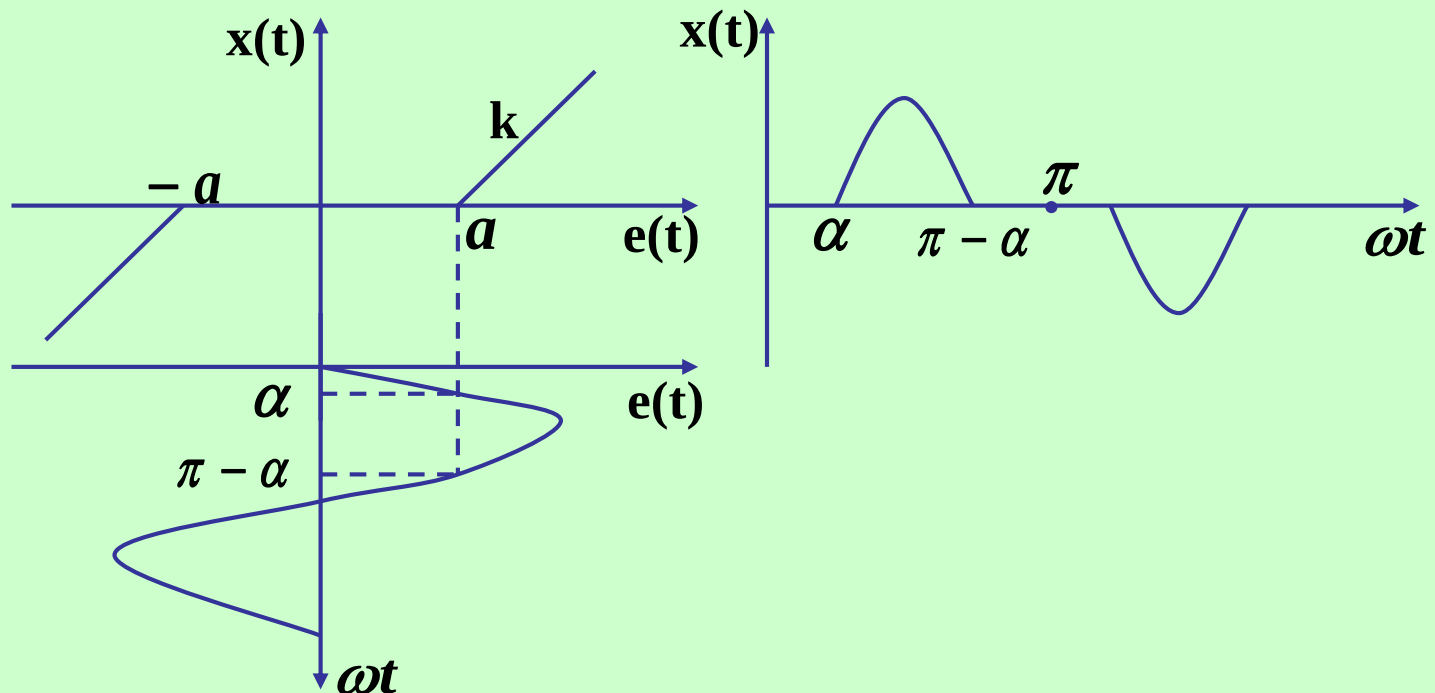


Fig8-11 input and output waveforms of dead zone nonlinearity

When $A > a$, the output of dead zone nonlinearity is

$$x(t) = \begin{cases} 0, & 0 \leq \omega t \leq \alpha \\ K(A \sin \omega t - a), & \alpha < \omega t \leq \pi - \alpha \\ 0, & \pi - \alpha < \omega t \leq \pi \end{cases}$$

where, $A \sin \alpha = a, \therefore \alpha = \sin^{-1} \frac{a}{A}$

The output is odd $\rightarrow A_1 = 0, \phi_1 = 0$

$$\begin{aligned} B_1 &= \frac{2}{\pi} \int_0^\pi x(t) \sin \omega t d(\omega t) \\ &= \frac{2}{\pi} \int_\alpha^{\pi-\alpha} K(A \sin \omega t - a) \sin \omega t d(\omega t) \end{aligned}$$

$$B_1 = \frac{2}{\pi} KA \left[\frac{\pi}{2} - \sin^{-1} \frac{a}{A} - \frac{a}{A} \sqrt{1 - \left(\frac{a}{A} \right)^2} \right]$$

The describing function of dead zone nonlinearity is

$$N(A) = \frac{B_1}{A} = \frac{2}{\pi} K \left[\frac{\pi}{2} - \sin^{-1} \frac{a}{A} - \frac{a}{A} \sqrt{1 - \left(\frac{a}{A} \right)^2} \right]$$

Note:

(1) When a / A is very small, i.e., the non-sensible zone is small, $N(A)$ is approximately equal to K ;

(2) $N(A)$ decreases as a / A becomes larger

(3) $a / A = 1 \quad \Rightarrow \quad N(A) = 0$

3. Gap

Assume the input is $e(t) = A \sin \omega t$,

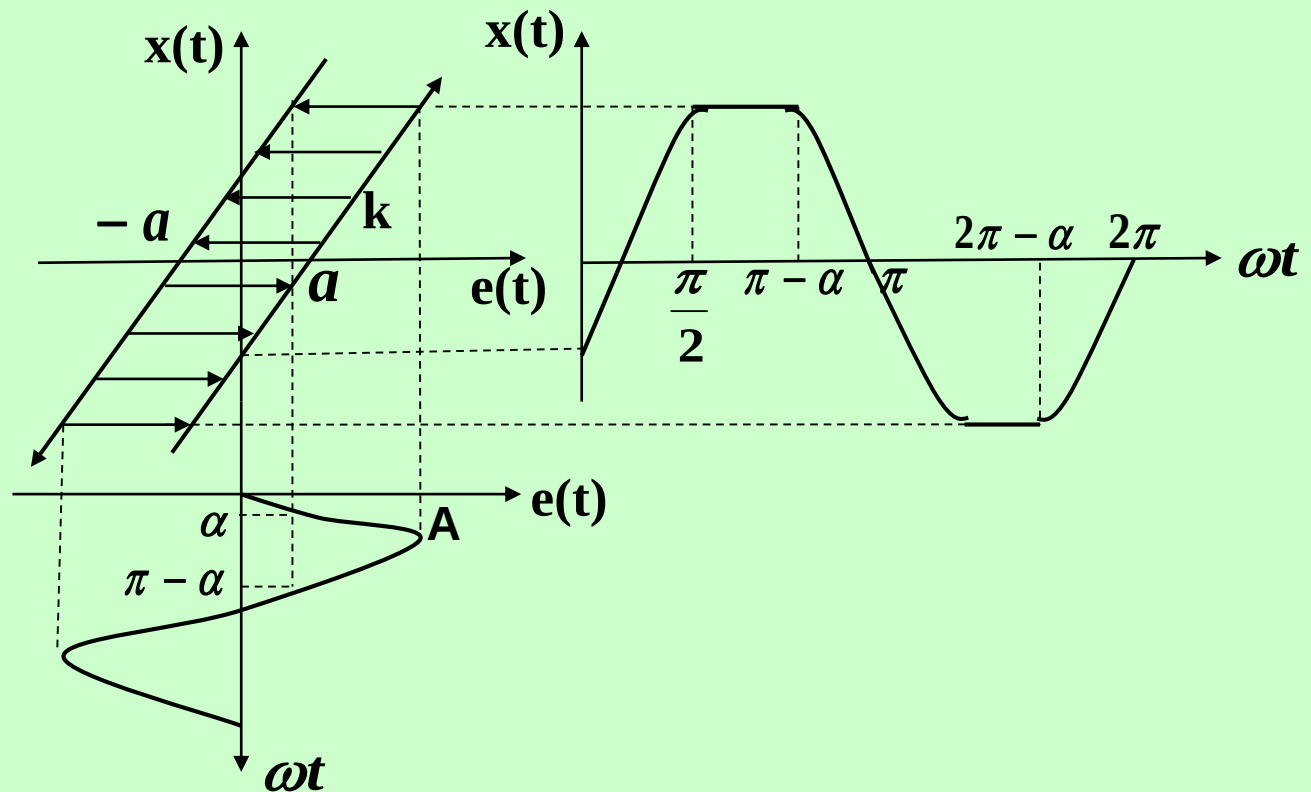
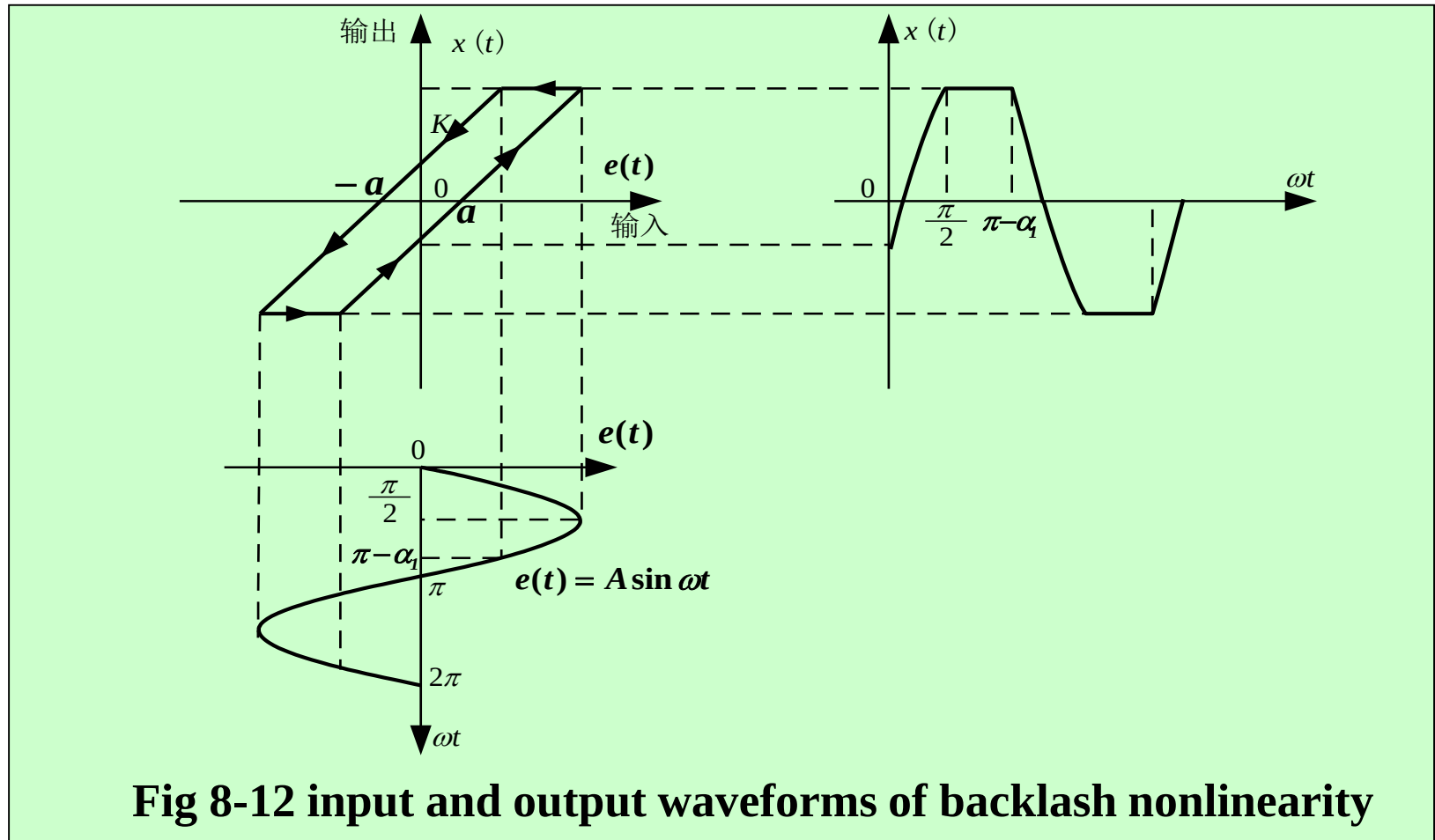


Fig 8-12 input and output waveforms of gap nonlinearity

Assume the input is $e(t) = A \sin \omega t$,



From the mathematical description of gap nonlinearity, the $x(t)$ is given by

$$x(t) = \begin{cases} K(A \sin \omega t - a), & 0 \leq \omega t < \frac{\pi}{2} \\ K(A - a), & \frac{\pi}{2} \leq \omega t < \pi - \alpha \\ K(A \sin \omega t + a), & \pi - \alpha \leq \omega t \leq \pi \end{cases}$$

where, $A \sin(\pi - \alpha) = A - 2a$, $\therefore \alpha = \sin^{-1} \frac{A - 2a}{A}$

$$\begin{aligned}
 A_1 &= \frac{2}{\pi} \int_0^{\pi} x(t) \cos \omega t \, d(\omega t) \\
 &= \frac{2}{\pi} \int_0^{\frac{\pi}{2}} K(A \sin \omega t - a) \cos \omega t \, d(\omega t) \\
 &\quad + \frac{2}{\pi} \int_{\frac{\pi}{2}}^{\pi-\alpha} K(A - a) \cos \omega t \, d(\omega t) \\
 &\quad + \frac{2}{\pi} \int_{\pi-\alpha}^{\pi} K(A \sin \omega t + a) \cos \omega t \, d(\omega t) = \frac{4KA}{\pi} \left[\left(\frac{a}{A} \right)^2 - \frac{a}{A} \right]
 \end{aligned}$$

$$\begin{aligned}
 B_1 &= \frac{2}{\pi} \int_0^{\pi} x(t) \sin \omega t \, d(\omega t) = \frac{2}{\pi} \int_0^{\frac{\pi}{2}} K(A \sin \omega t - a) \sin \omega t \, d(\omega t) \\
 &= \frac{KA}{\pi} \left[\frac{\pi}{2} + \sin^{-1} \left(\frac{A - 2a}{A} \right) + \frac{A - 2a}{A} \sqrt{1 - \left(\frac{A - 2a}{A} \right)^2} \right]
 \end{aligned}$$

So, we can obtain the describing function $N(A)$ of gap nonlinearity as follows:

$$\begin{aligned}
 N(A) &= \frac{B_1}{A} + j \frac{A_1}{A} \\
 &= \frac{K}{\pi} \left[\frac{\pi}{2} + \sin^{-1} \left(\frac{A-2a}{A} \right) + \frac{A-2a}{A} \sqrt{1 - \left(\frac{A-2a}{A} \right)^2} \right] + j \frac{4K}{\pi} \left[\frac{a(a-A)}{A^2} \right] \\
 &= |N(A)| e^{j\phi_1}
 \end{aligned}$$

$$|N(A)| = \sqrt{\left[\frac{4K}{\pi} \left(\frac{a(a-A)}{A^2} \right) \right]^2 + \left[\frac{K}{\pi} \left(\frac{\pi}{2} + \sin^{-1} \frac{A-2a}{A} + \frac{A-2a}{A} \sqrt{1 - \left(\frac{A-2a}{A} \right)^2} \right) \right]^2}$$

$$\phi_1 = \tan^{-1} \frac{4 \frac{a(a-A)}{A^2}}{\left[\frac{\pi}{2} + \sin^{-1} \left(\frac{A-2a}{A} \right) + \frac{A-2a}{A} \sqrt{1 - \left(\frac{A-2a}{A} \right)^2} \right]}$$

4. Relay

Assume the input is $e(t) = A \sin \omega t$,

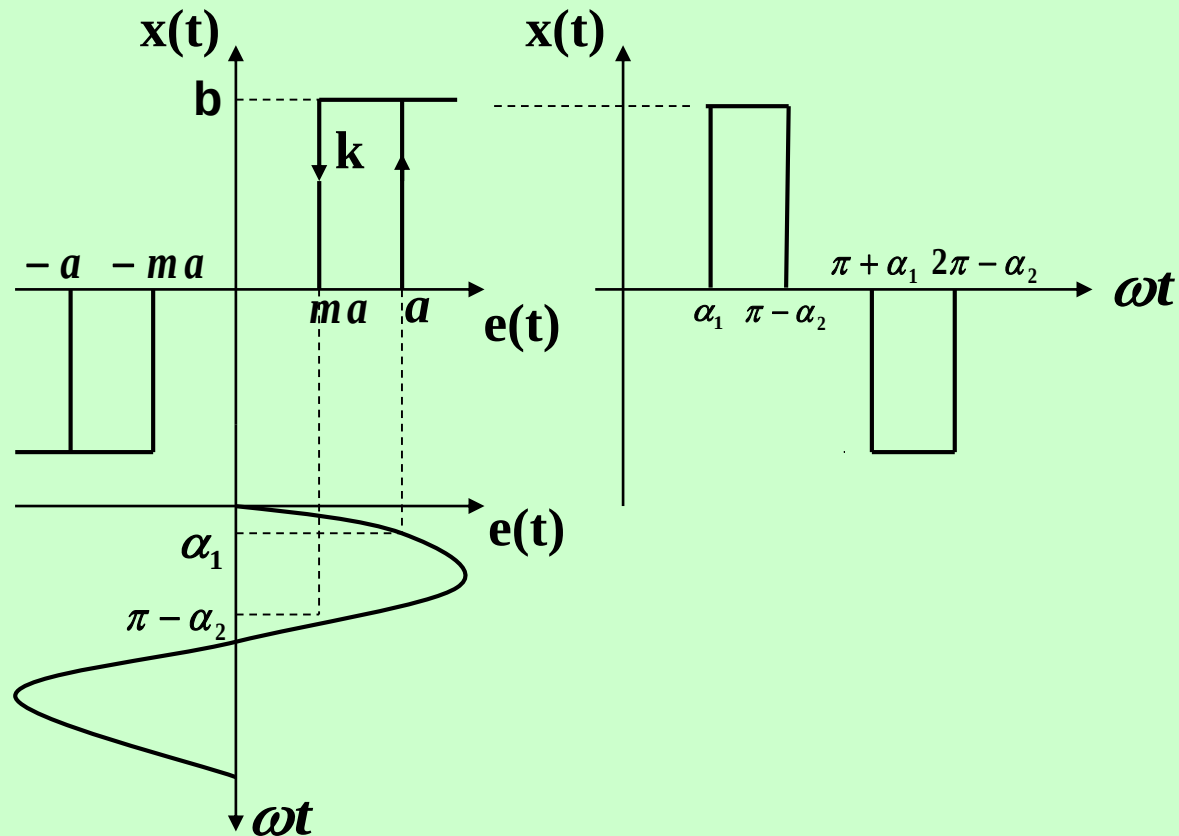


Fig 8-13 input and output waveforms of relay nonlinearity with dead zone and hysteresis ring

The output of the relay characteristic is :

$$x(t) = \begin{cases} 0, & 0 \leq \omega t < \alpha_1 \\ b, & \alpha_1 \leq \omega t < \pi - \alpha_2 \\ 0, & \pi - \alpha_2 \leq \omega t \leq \pi \end{cases}$$

where,

$$A \sin \alpha_1 = a, \therefore \alpha_1 = \sin^{-1} \frac{a}{A}$$

$$A \sin(\pi - \alpha_2) = ma, \therefore \alpha_2 = \sin^{-1} \frac{ma}{A}$$

$$A_1 = \frac{2}{\pi} \int_{\alpha_1}^{\pi - \alpha_2} b \cos \omega t d(\omega t)$$

$$= \frac{2b}{\pi} (\sin \alpha_2 - \sin \alpha_1) = \frac{2ab(m-1)}{\pi A}$$

$$B_1 = \frac{2}{\pi} \int_{\alpha_1}^{\pi - \alpha_2} b \sin \omega t d(\omega t)$$

$$= \frac{2b}{\pi} (\cos \alpha_2 + \cos \alpha_1) = \frac{2b}{\pi} \left[\sqrt{1 - \left(\frac{ma}{A} \right)^2} + \sqrt{1 - \left(\frac{a}{A} \right)^2} \right]$$

The describing function $N(A)$ of relay nonlinearity with dead zone and hysteresis ring is

$$N(A) = |N(A)|e^{j\phi_1} = \sqrt{\left(\frac{A_1}{A}\right)^2 + \left(\frac{B_1}{A}\right)^2} e^{jtg^{-1}\frac{A_1}{B_1}}$$

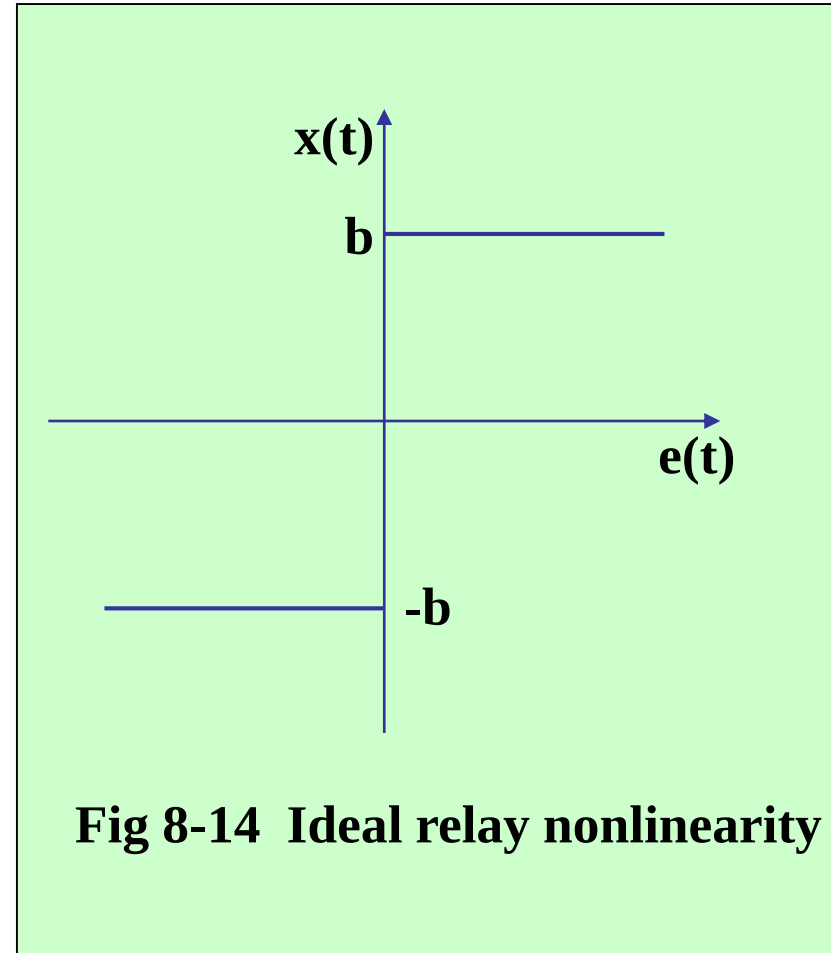
$$|N(A)| = \frac{2b}{\pi A} \sqrt{2 \left[1 - m \left(\frac{a}{A} \right)^2 + \sqrt{1 + m^2 \left(\frac{a}{A} \right)^4 - (m^2 + 1) \left(\frac{a}{A} \right)^2} \right]}$$

$$\phi_1 = tg^{-1} \frac{(m-1) \left(\frac{a}{A} \right)}{\sqrt{1 - m^2 \left(\frac{a}{A} \right)^2} + \sqrt{1 - \left(\frac{a}{A} \right)^2}}$$

Corollary:

When $a=0$, we can obtain the describing function of a *ideal relay nonlinearity*

$$N(A) = \frac{4b}{\pi A}$$



When $m=1$ and $a \neq 0$, we can obtain the describing function of a relay with dead zone

$$N(A) = \frac{4b}{\pi A} \sqrt{1 - \left(\frac{a}{A}\right)^2}$$

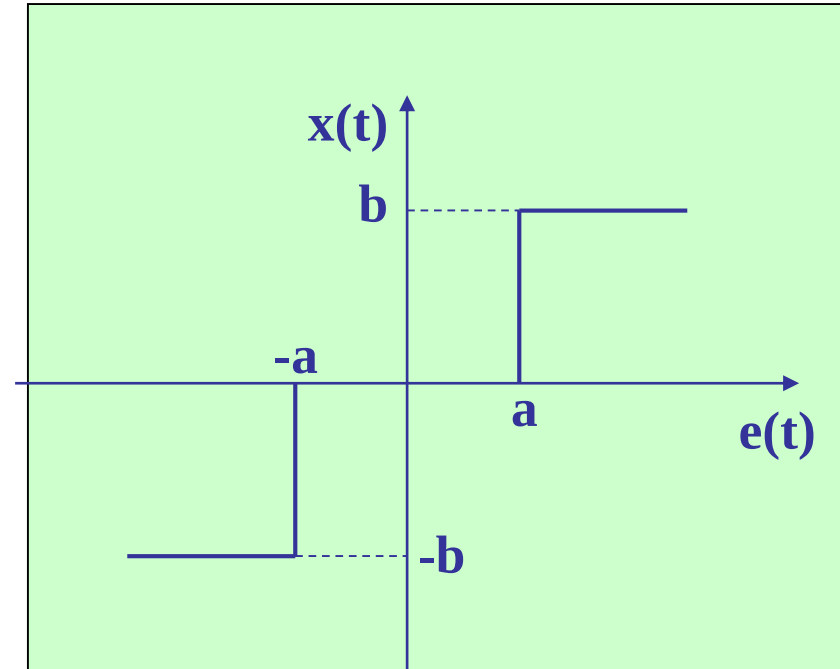


Fig 8-15 A relay with dead zone

When $m = -1$, we can obtain the describing function of a relay with hysteresis

$$N(A) = \frac{4b}{\pi A} e^{j\tau\omega} \frac{-\left(\frac{a}{A}\right)}{\sqrt{1 - \left(\frac{a}{A}\right)^2}}$$

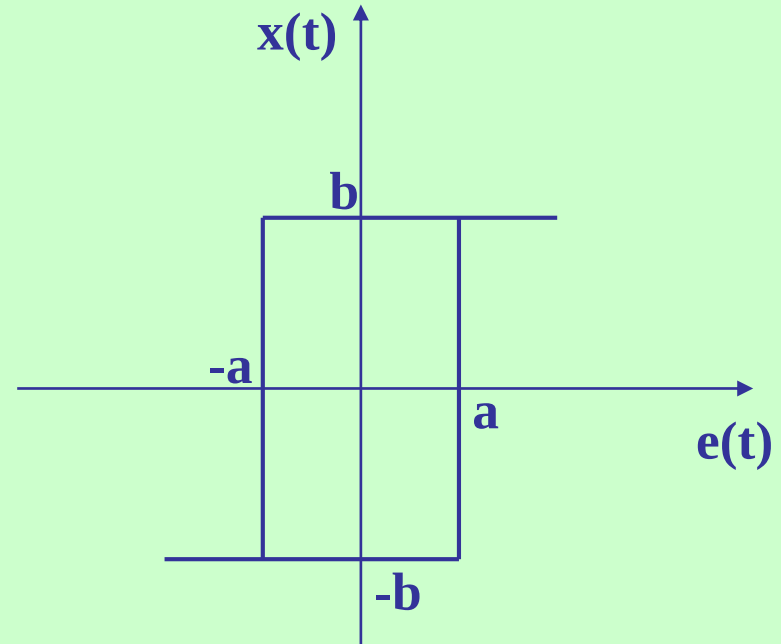


Fig 8-16 A relay with hysteresis

◆ 描述函数的求取

由描述函数的定义可以看出，求描述函数的步骤为：

- 1) 绘制输入—输出波形图，写出输入为正弦函数时非线性输出表达式；
- 2) 由波形图分析 $x(t)$ 的对称性，并由

$$A_1 = \frac{1}{\pi} \int_0^{2\pi} x(t) \cos \omega t d(\omega t) \quad B_1 = \frac{1}{\pi} \int_0^{2\pi} x(t) \sin \omega t d(\omega t)$$

计算 A_1 ， B_1 或者由

$$\varphi_1 = \text{tg}^{-1} A_1 / B_1 \quad X_1 = \sqrt{A_1^2 + B_1^2}$$

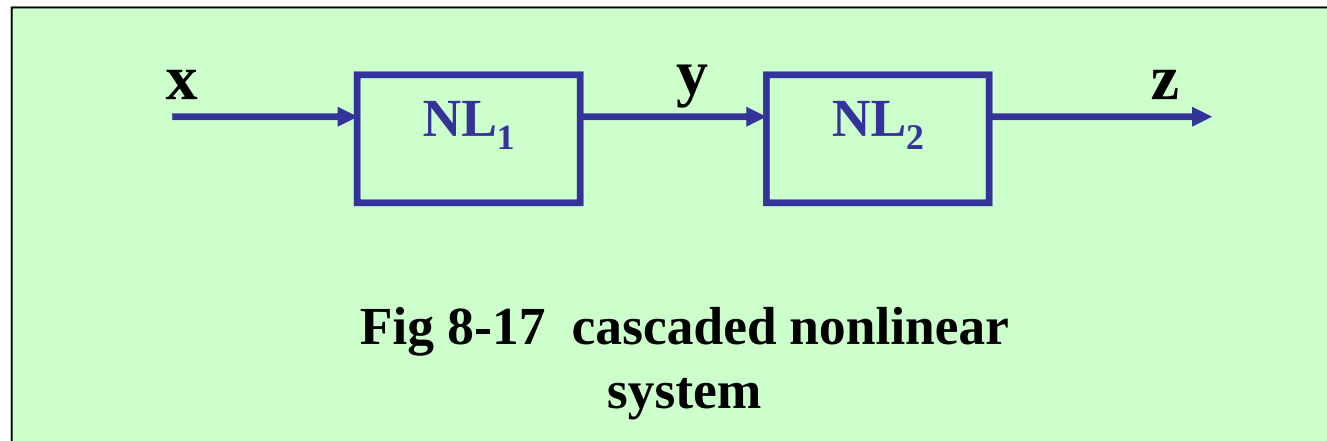
计算 X_1 ， φ_1 ；

- 3) 描述函数为

$$N(A) = \frac{B_1}{A} + j \frac{A_1}{A} = \frac{X_1}{A} e^{j\varphi_1}$$

8.3.3 Describing function of multiple nonlinearities

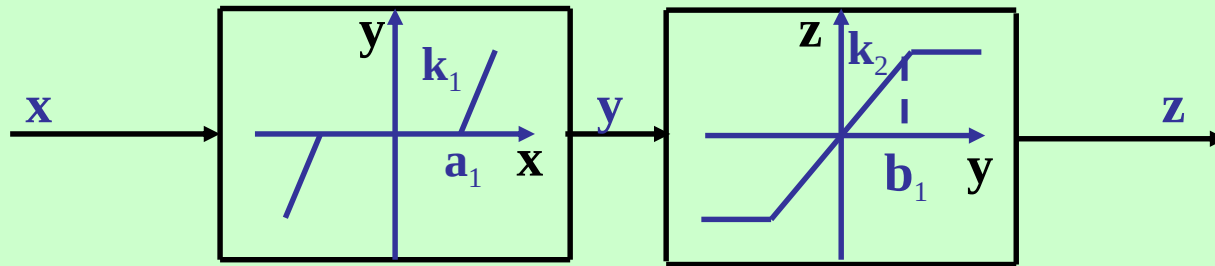
1. Cascaded nonlinear system



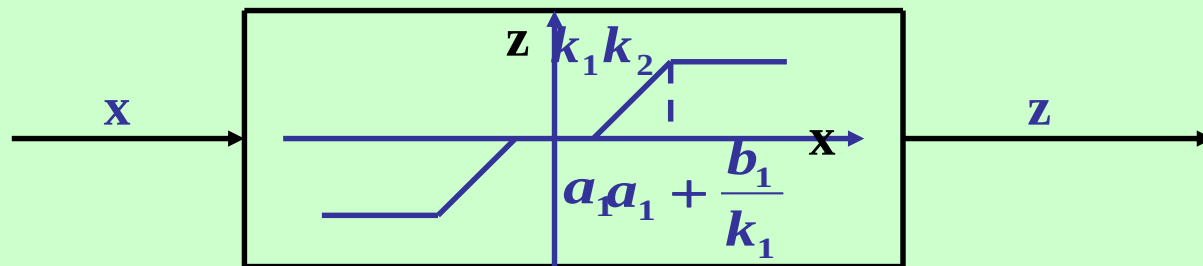
The describing function of a cascaded nonlinear system *is not equal to* the product of the two describing functions of each nonlinear elements.

$$N(A) \neq N_1(A) \cdot N_2(A)$$

assume NL_1 is a dead zone nonlinearity, NL_2 is a saturation nonlinearity, the composite nonlinearity of the cascaded nonlinear system is shown in Fig.8-18.



(a) cascaded nonlinear system



(b) composite nonlinearity

Fig 8-18 the cascaded nonlinear system and its composite nonlinearity

2. Parallel nonlinear system

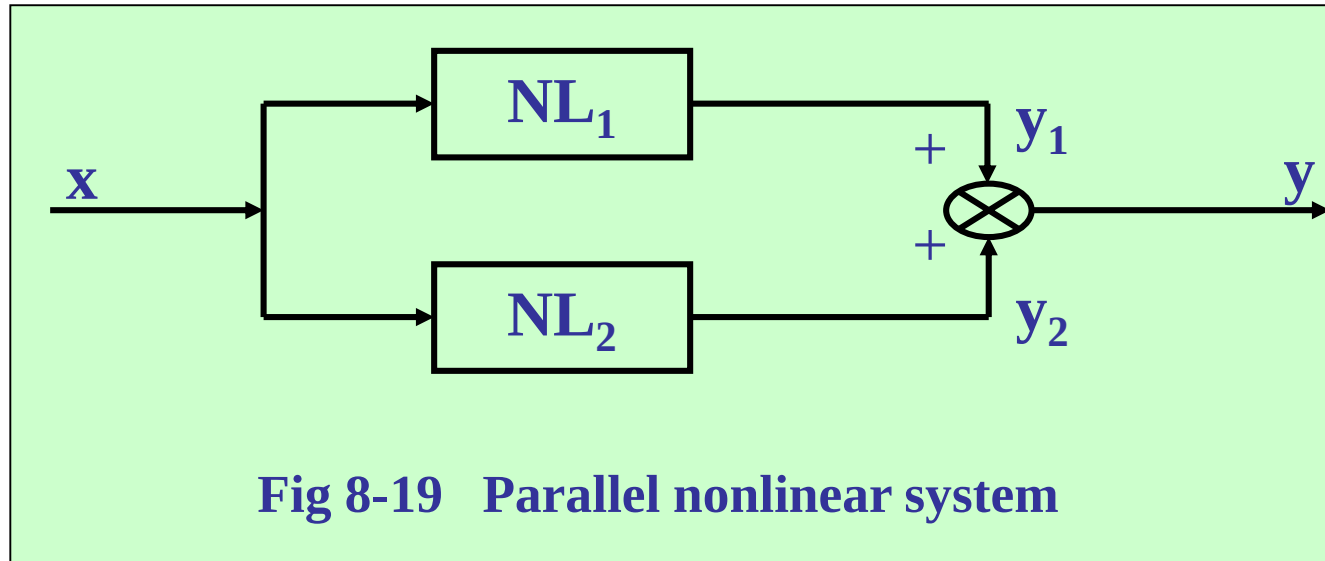


Fig 8-19 Parallel nonlinear system

According to the definition of describing function, the describing function $N(A)$ of the output y and the input x is equal to the sum of two describing functions:

$$N(A) = N_1(A) + N_2(A)$$

Example:

$$y = \frac{1}{2}x + \frac{1}{4}x^3$$

Determine the describing function of the system.

Solution: Since y is an odd function, A_1 is zero.

$$N = \frac{B_1}{A} = \frac{1}{\pi A} \int_0^{2\pi} \left(\frac{1}{2}x + \frac{1}{4}x^3 \right) \sin \omega t d(\omega t) \quad (1)$$

Let $x = A \sin \omega t$, Eq (1) becomes

$$N = \frac{1}{\pi A} \int_0^{2\pi} \left(\frac{1}{2}A \sin \omega t + \frac{1}{4}A^3 \sin^3 \omega t \right) \sin \omega t d(\omega t)$$

Let $x = A \sin \omega t$, Eq (1) becomes

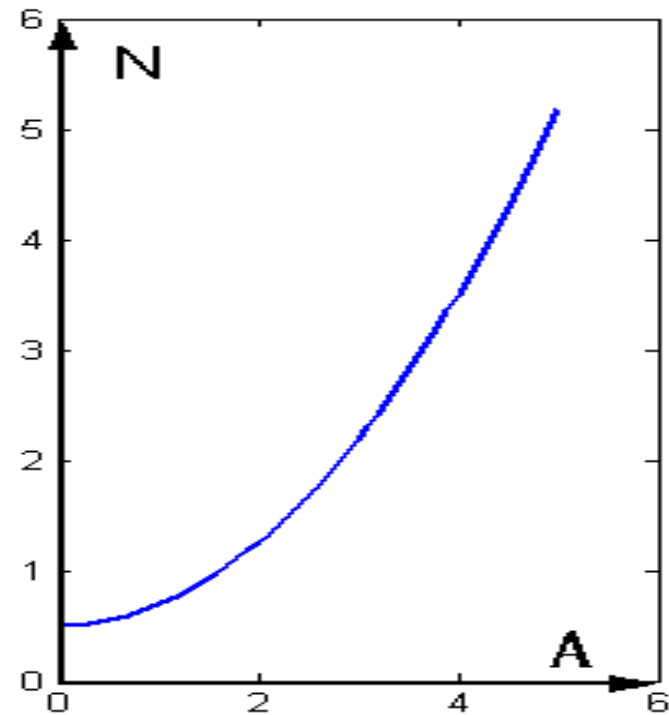
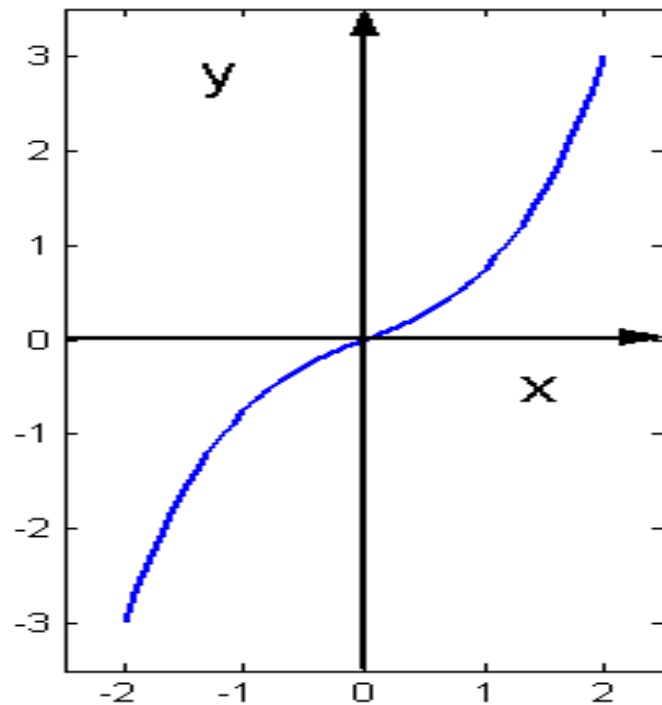
$$\begin{aligned} N &= \frac{1}{\pi A} \int_0^{2\pi} \left(\frac{1}{2} A \sin \omega t + \frac{1}{4} A^3 \sin^3 \omega t \right) \sin \omega t d(\omega t) \\ &= \frac{1}{\pi} \int_0^\pi \sin^2 \omega t d(\omega t) + \frac{A^2}{2\pi} \int_0^\pi \sin^4 \omega t d(\omega t) \end{aligned}$$

So

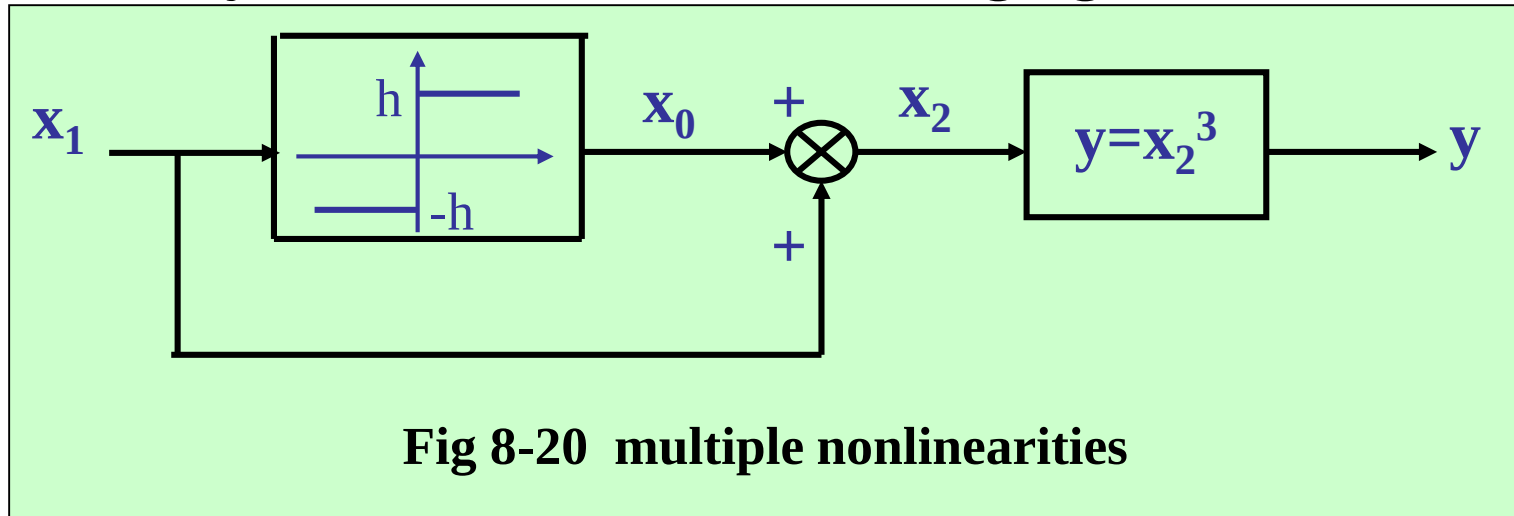
$$N = \frac{B_1}{A} = \frac{1}{2} + \frac{3}{16} A^2$$

$$y(t) = \left(\frac{1}{2} + \frac{3}{16} A^2 \right) A \sin \omega t$$

The out-input characteristics of nonlinearity



[Example 1] Obtain the describing function of the nonlinear system shown in the following figure.



Solution: $y = x_2^3 = (x_0 + x_1)^3 = x_0^3 + 3x_0^2x_1 + 3x_0x_1^2 + x_1^3$

then $N(A) = N_1(A) + N_2(A) + N_3(A) + N_4(A)$

assume $x_1 = A \sin \omega t$

NL₁ is an ideal relay nonlinearity when $a=0$,

$\therefore A_1 = 0$

Obtain $N_1(A)$:

$$B_1 = \frac{2}{\pi} \int_0^\pi h^3 \sin \omega t d(\omega t) = \frac{4h^3}{\pi}$$

$$\therefore N_1(A) = \frac{B_1}{A} = \frac{4h^3}{\pi A}$$

Obtain $N_2(A)$:

$$B_1 = \frac{2}{\pi} \int_0^\pi 3h^2 A \sin \omega t \cdot \sin \omega t d(\omega t) = 3h^2 A$$

$$\therefore N_2(A) = 3h^2$$

Obtain $N_3(A)$:

$$B_1 = \frac{2}{\pi} \int_0^\pi 3hA^2 \sin^2 \omega t \cdot \sin \omega t d(\omega t) = \frac{8hA^2}{\pi}$$

$$\therefore N_3(A) = \frac{8hA}{\pi}$$

Obtain $N_4(A)$:

$$\begin{aligned} B_1 &= \frac{2}{\pi} \int_0^\pi A^3 \sin^3 \omega t \cdot \sin \omega t d(\omega t) \quad \text{suppose } \theta = \omega t \\ &= \frac{2}{\pi} \int_0^\pi -A^3 \sin^3 \theta \cdot d(\cos \theta) \\ &= \frac{2A^3}{\pi} \left[\left(-\sin^3 \theta \cos \theta \right)_0^\pi + \int_0^\pi 3 \sin^2 \theta \cos^2 \theta d\theta \right] = \frac{3}{4} A^3 \end{aligned}$$

$$\therefore N_4(A) = \frac{3}{4} A^2$$

Then, the describing function of the multiple nonlinearity is

$$N(A) = \frac{4h^3}{\pi A} + 3h^2 + \frac{8hA}{\pi} + \frac{3}{4} A^2$$

8.3.4 Analyze nonlinear system with describing function method

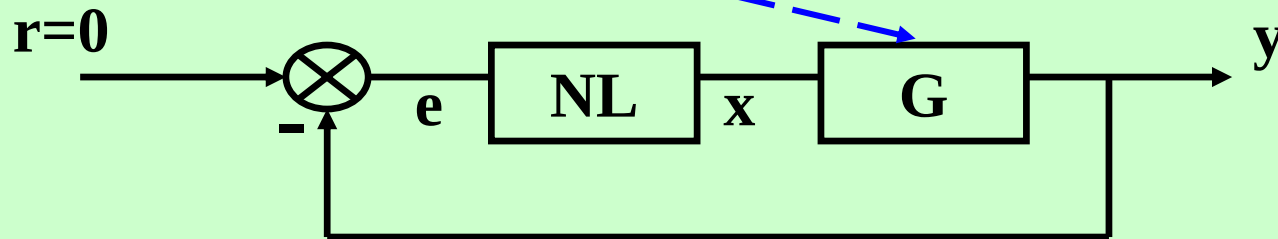
$$N(A) = \frac{\text{Fundamental component of the output } x(t)}{\text{Amplitude of the sinusoidal input } e(t)}$$

➡ It can only reflect *partial dynamic characteristics* of the system.

If a *self-excited oscillation* occurs in the system, any input $x(t)$ can be regarded as a *sinusoidal* signal because the linear part is a *low-pass filter*.



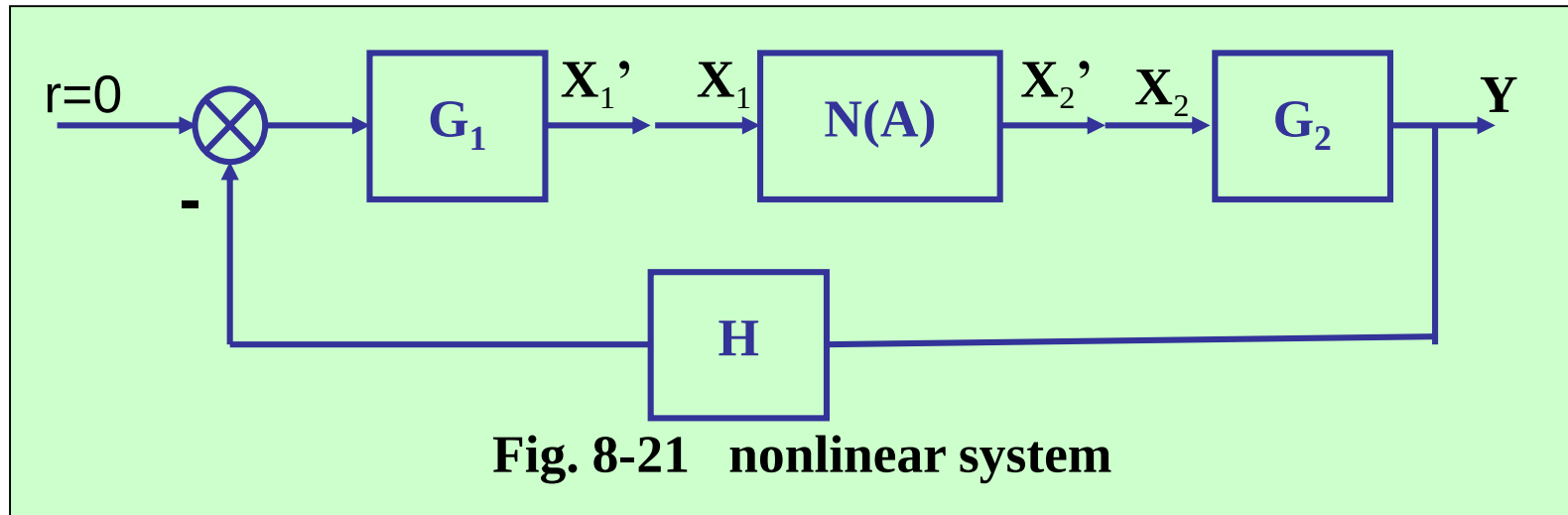
The describing function method is then *applicable*.



Typical structure of nonlinear system

Assume the input of a nonlinear system is zero,

$N(A)$ is the describing function of a nonlinear element,
analyze the *condition for self-excited oscillation*.



Assume $X_2 = A_2 \sin \omega t$

Then $X_1' = -|G_1(j\omega)G_2(j\omega)H(j\omega)|A_2 \sin(\omega t + \theta)$

where: $\theta = \angle G_1(j\omega) + \angle G_2(j\omega) + \angle H(j\omega)$

Assume: $N(A) = |N(A)|e^{j\phi}$

then $x_2'(t) = -|N(A)||G_1(j\omega)G_2(j\omega)H(j\omega)|A_2 \sin(\omega t + \theta + \phi)$

Note: $x_2'(t) = x_2(t) \iff$ **The self-oscillation occurs.**

Considering we have $x_2 = A_2 \sin \omega t$,
the *condition of self-oscillation* is

$$\left\{ \begin{array}{l} |N(A)||G_1(j\omega)G_2(j\omega)H(j\omega)| = 1 \\ \theta + \phi = (2n + 1)\pi \end{array} \right.$$

$$\left\{ \begin{array}{l} |N(A)| |G_1(j\omega)G_2(j\omega)H(j\omega)| = 1 \\ \theta + \phi = (2n + 1)\pi \end{array} \right.$$

Suppose the transfer function of linear part satisfies

$$G(s) = G_1(s)G_2(s)H(s)$$

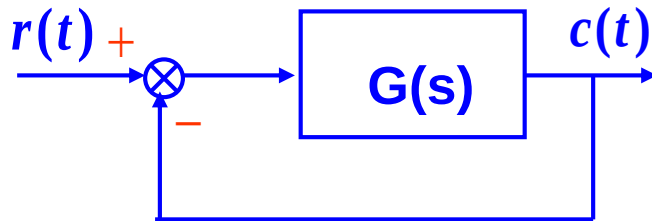
We can conclude that the *condition of self-excited oscillation* is

$$G(j\omega) = -\frac{1}{N(A)}$$

or $1 + N(A)G(j\omega) = 0$

Review of Nyquist criterion

For the linear system:



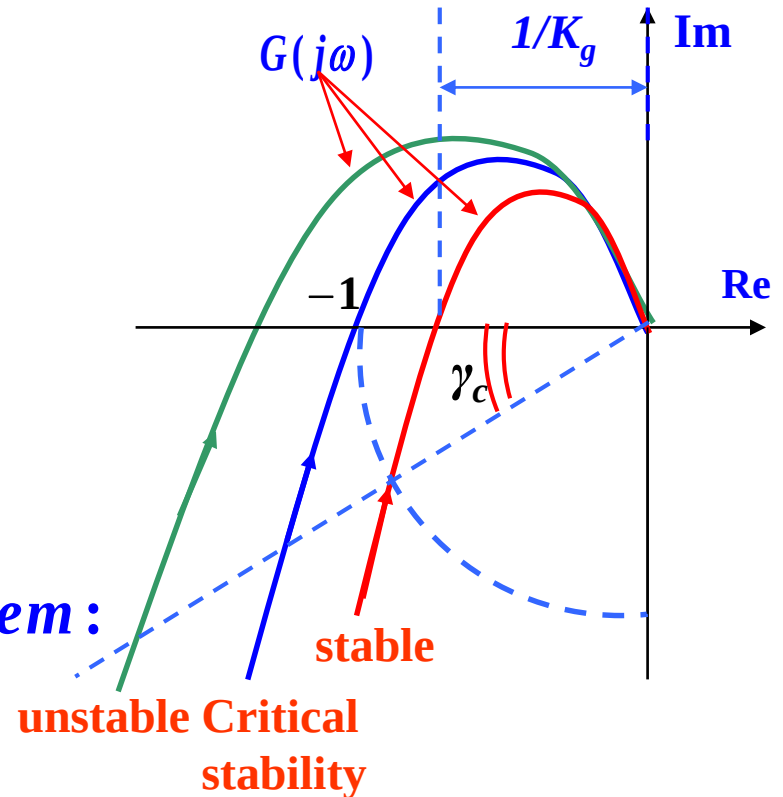
The characteristic equation of the system:

$$1 + G(j\omega) = 0$$

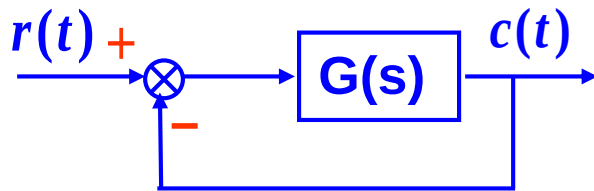
$$\Rightarrow G(j\omega) = -1 + j0$$

If $G(s)$ is a minimum phase transfer function, the necessary and sufficient condition of the stable system is:

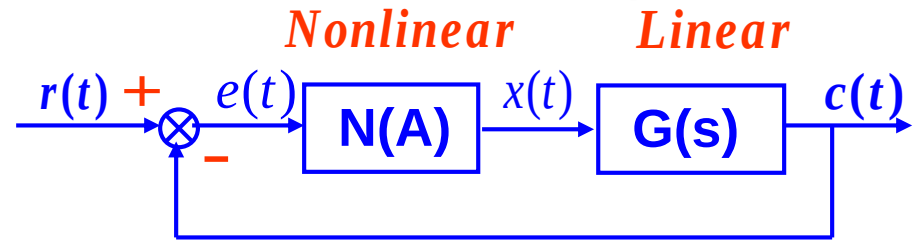
$G(j\omega)$ does not circle the point $(-1, j0)$



Compare the nonlinear system with the linear system



Linear system



nonlinear system

Transfer function of the system:

$$\phi(j\omega) = \frac{C(j\omega)}{R(j\omega)} = \frac{G(j\omega)}{1 + G(j\omega)}$$

$$\phi(j\omega) = \frac{C(j\omega)}{R(j\omega)} = \frac{N(A)G(j\omega)}{1 + N(A)G(j\omega)}$$

Characteristic equation:

$$1 + G(j\omega) = 0$$

$$\Rightarrow G(j\omega) = -1$$

In the $G(j\omega)$ plane

A point

$$1 + N(A)G(j\omega) = 0$$

$$\Rightarrow G(j\omega) = -\frac{1}{N(A)}$$

A curve

Because the describing function $N(A)$ actually is a linearized frequency response, we can expand the Nyquist criterion to the nonlinear system:

Stability analysis of the nonlinear system

(For example the minimum phase system)

*compare with
linear system*

(1) $G(j\omega)$ don't circle the $-\frac{1}{N(A)}$ curve, the nonlinear system is stable

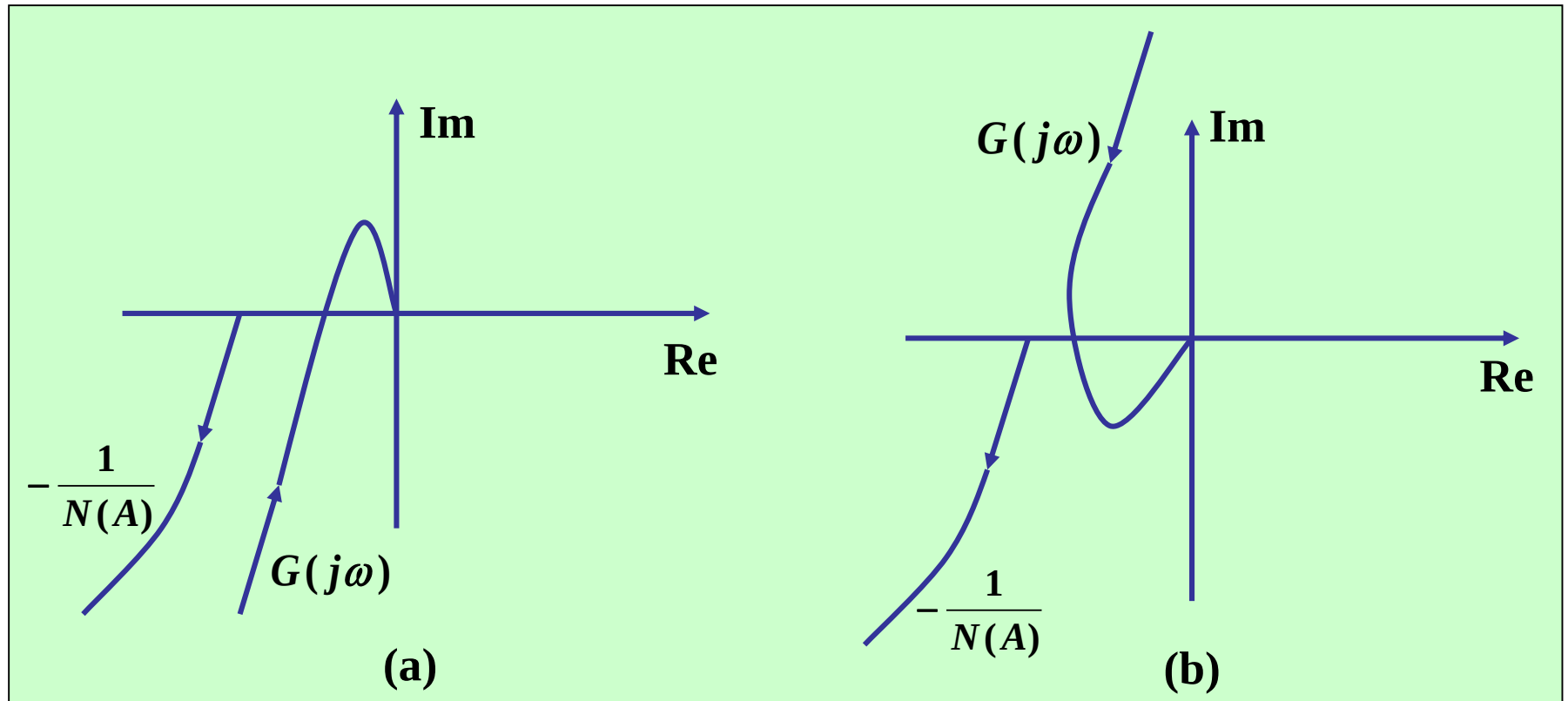
(2) $G(j\omega)$ circle the $-\frac{1}{N(A)}$ curve, the nonlinear system is unstable;

(3) $G(j\omega)$ intersect with the $-\frac{1}{N(A)}$ curve, there will be a self-oscillation in the nonlinear system .

(1) $G(j\omega)$ don't circle the point $(-1, j\omega)$, the system is stable;

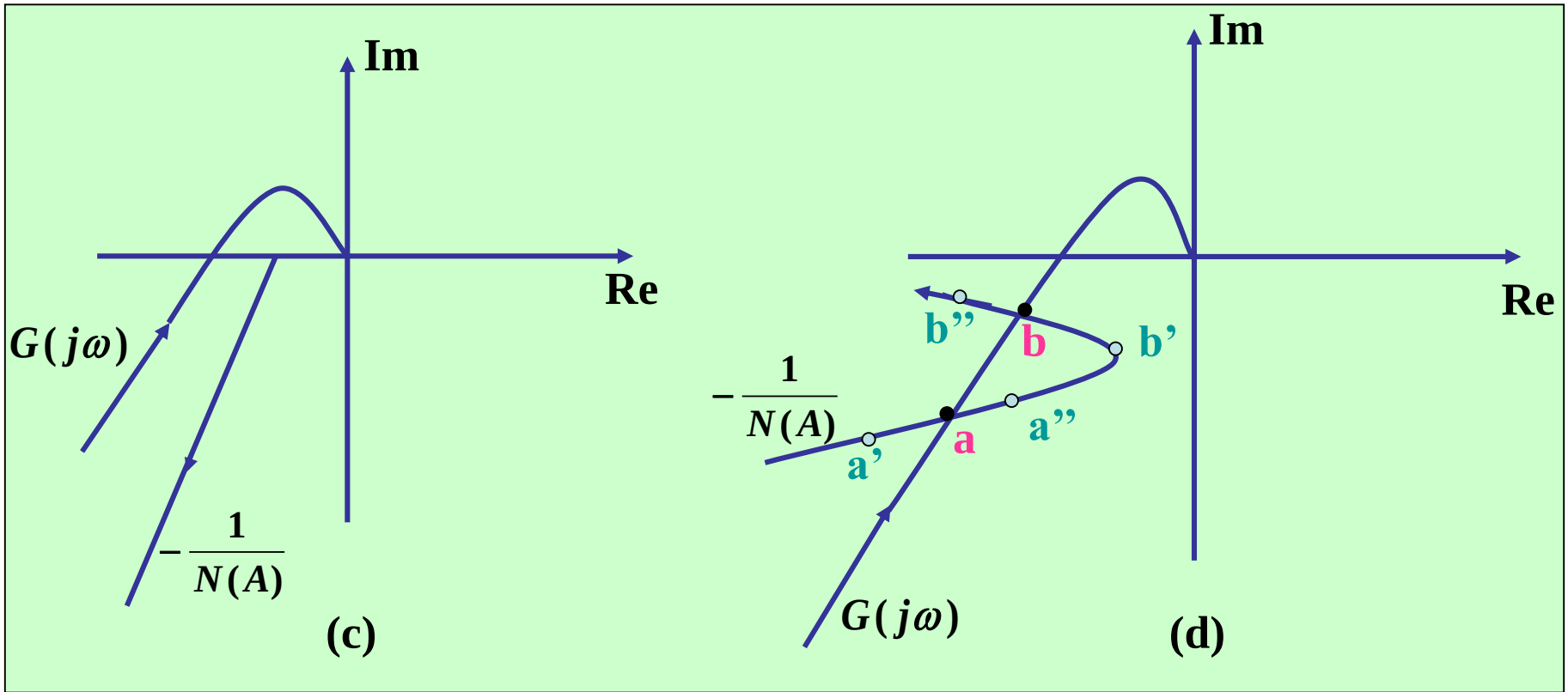
(2) $G(j\omega)$ circle the point $(-1, j\omega)$, the system is unstable;

(3) $G(j\omega)$ intersect with the point $(-1, j\omega)$, the system is in the critical stability.



$G(j\omega)$ do not circle $-\frac{1}{N(A)}$
(Stable)

$G(j\omega)$ circle $-\frac{1}{N(A)}$
(Unstable)

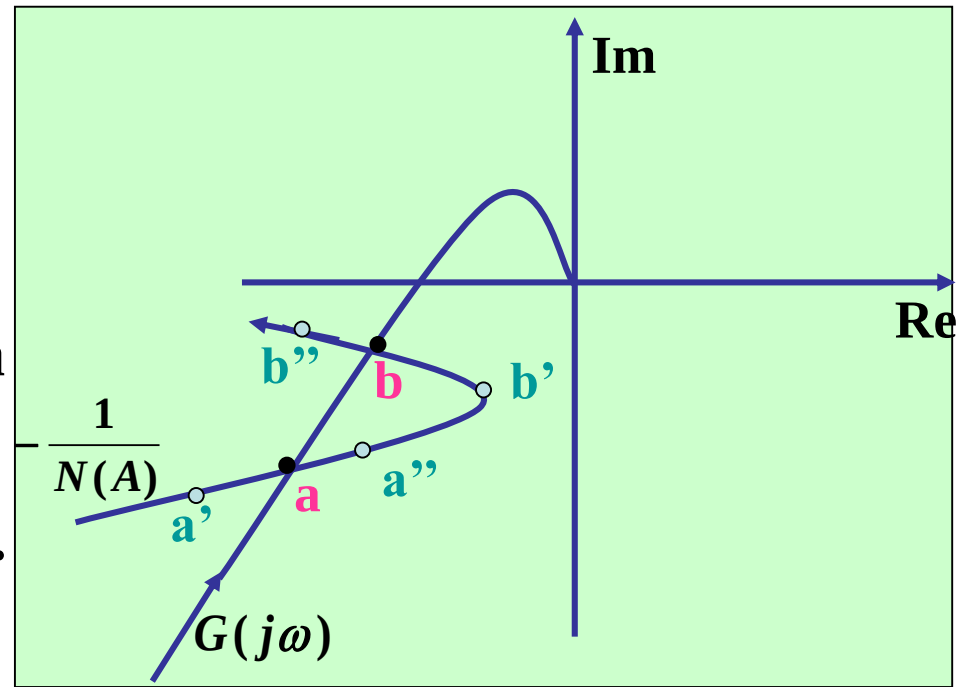


Unstable

$G(j\omega)$ intersect with $-\frac{1}{N(A)}$
(Self-oscillation)

Analysis:

- (1) Self-oscillation occurs at point a and point b.
- (2) The self-oscillation at point a is unstable while the self-oscillation at point b is stable.
- (3) There is only one stable self-oscillation in an actual physical system.



The *amplitude* and *frequency* of the self-oscillation can be obtained by solving

$$|G(j\omega)N(A)| = 1$$

$$\theta + \phi = -\pi$$

[Example 2] A relay control system structure is shown in Fig. 8-23. Suppose $a=1$, $b=3$.

(1) Is there a self-oscillation in the system? If there is, obtain the amplitude and frequency of the oscillation.

(2) How to adjust the parameters if you want to eliminate the self-oscillation?

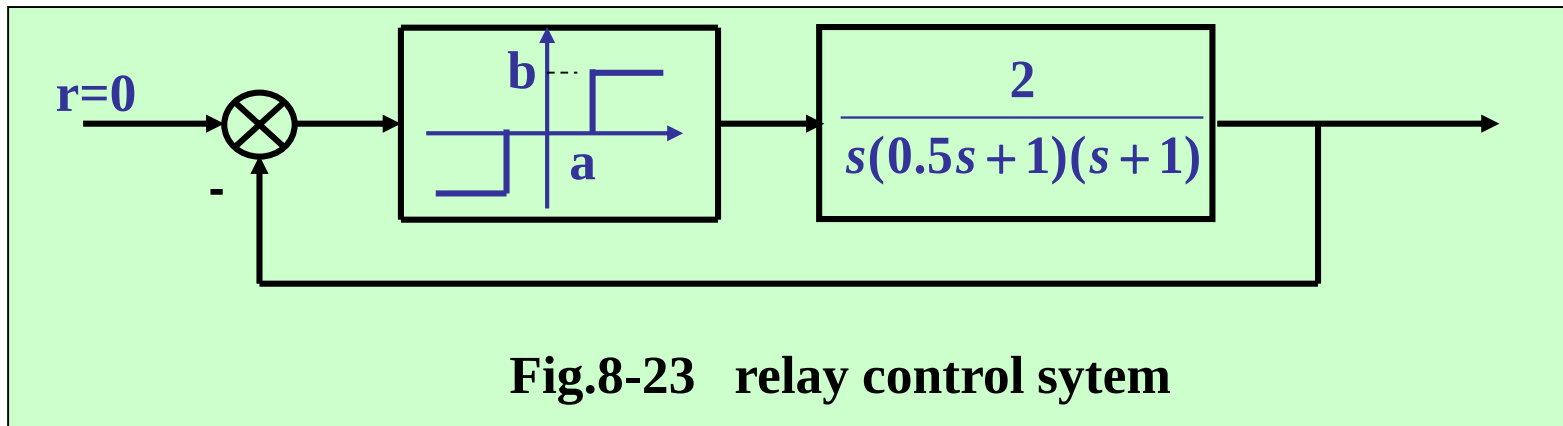


Fig.8-23 relay control sytem

Solution: the describing function of a relay nonlinearity with dead zone is

$$N(A) = \frac{4b}{\pi A} \sqrt{1 - \left(\frac{a}{A}\right)^2}$$

$$\therefore -\frac{1}{N(A)} = -\frac{\pi A}{4b\sqrt{1-\left(\frac{a}{A}\right)^2}}$$

when $A = a, -\frac{1}{N(A)} \rightarrow -\infty$

when $A \rightarrow \infty, -\frac{1}{N(A)} \rightarrow -\infty$

There is an **extreme value** of function $-\frac{1}{N(A)}$ on the real axis.

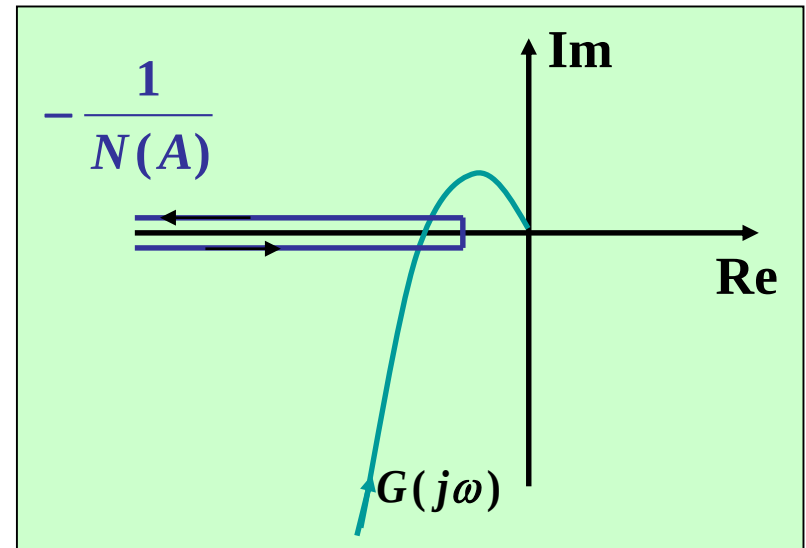
$$\frac{d}{dA}\left(-\frac{1}{N(A)}\right) = 0 \implies 1 - 2\left(\frac{a}{A}\right)^2 = 0 \quad \therefore A = \sqrt{2}a$$

Substitute $a=1$, $b=3$ into above equations, we have

$$A = \sqrt{2}$$

$$-\left. \frac{1}{N(A)} \right|_{A=\sqrt{2}} = -\frac{\pi}{6} \approx -0.52$$

$$G(s) = \frac{2}{s(0.5s + 1)(s + 1)}$$



$$G(j\omega) = -\frac{3\omega}{\omega(0.25\omega^4 + 1.25\omega^2 + 1)} - j\frac{2(1 - 0.5\omega^2)}{\omega(0.25\omega^4 + 1.25\omega^2 + 1)}$$

Set the imaginary part to zero, we have $\omega = \sqrt{2}$

Substituting $\omega = \sqrt{2}$ into the real part, we have

$$\operatorname{Re} G(j\omega) \Big|_{\omega=\sqrt{2}} = -\frac{1}{1.5} \approx -0.66$$

$$\text{Let } -\frac{1}{N(A)} = \frac{-\pi A}{12\sqrt{1-\left(\frac{1}{A}\right)^2}} = -\frac{1}{1.5}$$

We can obtain two amplitude: $A_1=1.11$, $A_2=2.3$


(not exist in the reality)

There is a self-oscillation in the system with the amplitude 2.3 and the frequency $\sqrt{2}$.

(2) To eliminate the self-oscillation, let

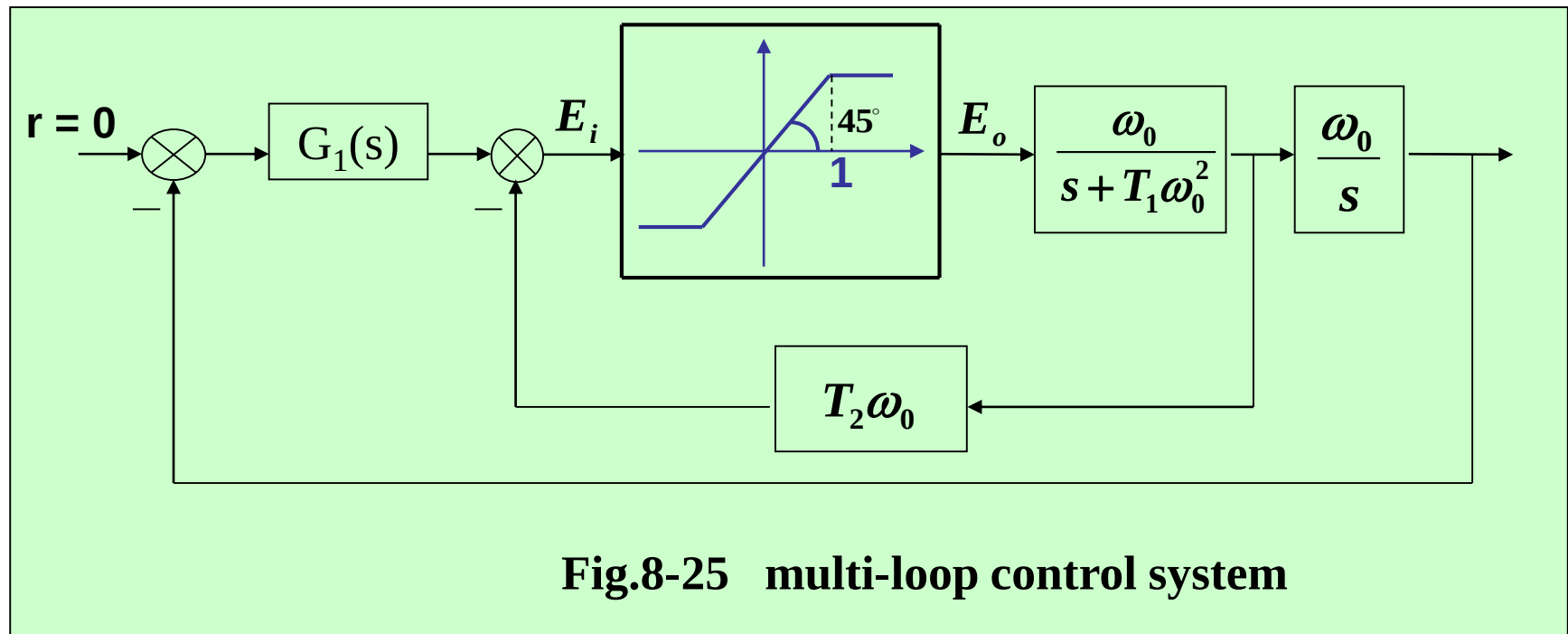
$$\left. -\frac{1}{N(A)} = -\frac{\pi A}{4b\sqrt{1-\left(\frac{a}{A}\right)^2}} \right|_{A=\sqrt{2}a} \leq -\frac{1}{1.5}$$

We can get the ratio of relay parameters

$$\frac{b}{a} \leq \frac{1.5\pi}{2} \approx 2.36$$

Adjust the ratio of a and b to $b=2a$, we can eliminate the self-oscillation.

[Example 3] A multi-loop control system is shown in Fig. 8-25. If $G_1(s)=1$, the **natural oscillation frequency** and **damping ratio** of system is $\omega_n = 2$ and $\zeta = 1$ when working on the linear area of the saturation. Suppose $G_1(s) = 1 + \frac{1}{8s}$, try to find the minimal ratio $\frac{T_1}{T_2}$ when the system is stable.



Solution: When $G_1(s)=1$, the closed-loop transfer function of the inner loop is:

$$G_{\text{内}}(s) = \frac{\frac{\omega_0}{s + T_1 \omega_0^2}}{1 + \frac{T_2 \omega_0^2}{s + T_1 \omega_0^2}} = \frac{\omega_0}{s + (T_1 + T_2) \omega_0^2}$$

The closed-loop transfer function of the whole system is

$$G_B(s) = \frac{\frac{\omega_0^2}{s[s + (T_1 + T_2) \omega_0^2]}}{1 + \frac{\omega_0^2}{s[s + (T_1 + T_2) \omega_0^2]}} = \frac{\omega_0^2}{s^2 + (T_1 + T_2) \omega_0^2 s + \omega_0^2}$$

$$\therefore \begin{cases} \omega_0 = \omega_n = 2 \\ T_1 + T_2 = \zeta = 1 \end{cases}$$

when $G_1(s) = 1 + \frac{1}{8s}$, the transfer function of inner loop is

$$G_{\text{内}}(s) = \frac{\frac{\omega_0}{s + T_1 \omega_0^2} N(A)}{1 + \frac{T_2 \omega_0^2}{s + T_1 \omega_0^2} N(A)} = \frac{\omega_0 N(A)}{s + T_1 \omega_0^2 + T_2 \omega_0^2 N(A)}$$

The open-loop transfer function of the whole system

$$G(s) = G_1(s)G_{\text{内}}(s) \frac{\omega_0}{s} = \frac{\omega_0^2 (1 + \frac{1}{8s}) N(A)}{s[s + T_1 \omega_0^2 + T_2 \omega_0^2 N(A)]}$$

From the characteristic equation of closed-loop system

$1 + G(s) = 0$, we have

$$s^2 + T_1\omega_0^2s + T_2\omega_0^2sN(A) + \omega_0^2\left(1 + \frac{1}{8s}\right)N(A) = 0$$

Substituting $\omega_0 = 2$ to the above equation, we can obtain

$$8s^3 + 32T_1s^2 + (32T_2s^2 + 32s + 4)N(A) = 0$$

$$\therefore -\frac{1}{N(A)} = \frac{8T_2s^2 + 8s + 1}{2s^3 + 8T_1s^2} = \frac{8T_2s^2 + 8s + 1}{s^2(2s + 8T_1)}$$

Note that the describing function of saturation nonlinearity is

$$N(A) = \frac{2}{\pi} \left[\sin^{-1} \frac{1}{A} + \frac{1}{A} \sqrt{1 - \left(\frac{1}{A}\right)^2} \right]$$

So the function of $-\frac{1}{N(A)}$ is given by

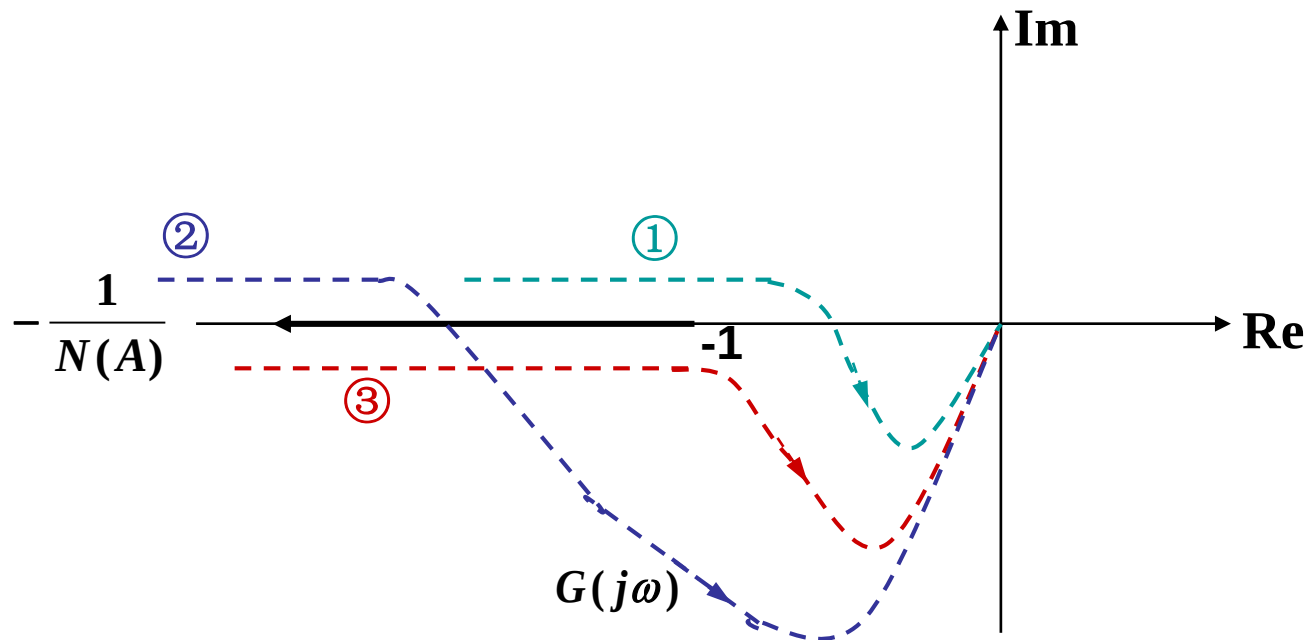
$$-\frac{1}{N(A)} = -\frac{\pi}{2} \frac{1}{\left[\sin^{-1} \frac{1}{A} + \frac{1}{A} \sqrt{1 - \left(\frac{1}{A} \right)^2} \right]}$$

The open-loop transfer function of the whole system is

$$G(s) = -\frac{1}{N(A)} = \frac{8T_2s^2 + 8s + 1}{s^2(2s + 8T_1)}$$

$$\therefore G(j\omega) = \frac{-8T_2\omega^2 + j8\omega + 1}{-\omega^2(j2\omega + 8T_1)} = u(\omega) + jv(\omega)$$

$$G(j\omega) = -\frac{4T_1 - 32\omega^2 T_1 T_2 + 8\omega^2}{2\omega^2(\omega^2 + 16T_1^2)} + j \frac{32T_1 - 1 + 8T_2\omega^2}{2\omega(\omega^2 + 16T_1^2)}$$



To guarantee the system to be stable, we should choose parameters T_1 and T_2 which make $G(j\omega)$ has no intersection with the negative real axis. So we choose curve ③

Let
$$v(\omega) = \frac{32T_1 - 1 + 8T_2\omega^2}{2\omega(\omega^2 + 16T_1^2)} = 0$$

Then we have
$$\omega = \sqrt{\frac{1 - 32T_1}{8T_2}}$$

If $T_1 > \frac{1}{32}$ holds, $G(j\omega)$ has no intersection with the negative real axis.

On the other hand, $T_1 + T_2 = 1$

$$\therefore T_2 < \frac{31}{32} \quad \Rightarrow \quad \therefore \frac{T_1}{T_2} \geq \frac{1}{31}$$

So, the minimal ratio of $\frac{T_1}{T_2}$ to guarantee the system to be

stable is
$$\frac{T_1}{T_2} = \frac{1}{31}$$